6SENG010C · SOFTWARE ENGINEERING FINAL PROJECT · PROJECT PROPOSAL

Market-State-Conditioned Orchestration of Heterogeneous Agents for Financial Decision Support

How should the influence of methodologically unlike financial agents be set when the reliability of their evidence channels varies with market conditions? I design, implement and evaluate a market-state-conditioned orchestration operator on Colombo Stock Exchange data.

STUDENT	IIT / UOW ID	SUPERVISOR	ACADEMIC YEAR
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PROPOSAL STORYLINE

A research spine and a software engineering spine, carried together

Two spines The left column is the academic inquiry; the right is the software engineering artefact.

Examiner focus Examiners will see concrete software engineering evidence backing each research assertion.

Table 1.1 · Dual-Spine Matrix

Research spine	Software engineering spine	Where it is
Agent reliability varies with market state; reviewed systems combine as though it does not	A pipeline that does not emit confident output from an empty channel	Slides 3, 9
59 reviewed papers, the six nearest systems contrasted clause by clause	Every requirement carries the paper or the measurement that forced it	Slides 4, 5, 6, 8, 10
One main question, two supporting, each answerable by a planned experiment	Four agents, one orchestrator, nine stated exclusions	Slides 6, 7, 10
Design science, controlled comparison, five arms, one ablation per state input	Five layers, point-in-time access, a verified data pipeline	Slides 11 to 15
An operator with a named lineage, plus evidence on whether it earns its place	Research prototype, no interface, three decisions changed by responsible practice	Slides 16 to 20

Spoken: I will show the gap and the contribution, then how the artefact, methodology and evaluation address it, then what I am not claiming. Slide 2

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PROBLEM AND GOAL · CRITERION 1, 10 PER CENT

Influence over a shared decision is fixed at design time

Four agents produce signals, and something must turn them into one decision. In the systems I reviewed that combination is set once and never moves, and in most of them there is no weight object at all to move.

SIGNIFICANCE

Evidence of significance

Three documented failures: a CSE study fixing a 3:1 split on an easier task's score; an optimiser fed uncalibrated model-authored returns; a pool degrading 10.36 points through a fixed prompt.

STAKEHOLDERS

Who is affected

The investment decision-maker acting on the output, the researcher running the comparison, and retail participants least able to absorb a cost a model assumed away.

GOAL

Project goal

Build the operator and produce attributable evidence on whether conditioning earns its place against fixed, static learned and performance-reactive weighting. Not on whether the system is profitable.

OUTCOMES

Expected outcomes

A pipeline where changing liquidity changes the weights, a five-arm comparison with ablation deltas, a CSE baseline set, and a stated position on confidence comparability.

Researching an orchestration mechanism, not building a profitable trading application.

Core premise

Four agents produce signals, but static combinations fail when individual channel reliability fluctuates.

Target

Attributable orchestration evidence, not portfolio profitability.

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Four lines of evidence, one gap they converge on

Table 2.1 · Gap Convergence

Evidence source	What it shows	Limitation or tension	Resulting gap
Academic literature, 59 papers	Heterogeneity is widely adopted; combination is treated as a solved detail	Several systems carry no weight object. Those that do weight assets, not influence	Influence is fixed at design time and adapts to nothing measured outside the agent
Existing systems	Four named systems each get one necessary clause right	The nearest neighbour in each direction is strong, so the contrast must be clause by clause	I have not found a system that combines all four clauses at once
Domain evidence, the CSE	Volume Granger-causes the index at p = 0.0041; macro variables reach nothing	Channels go empty for days, so equal trust across agents is false here	A setting where channel reliability genuinely varies, and can be measured
Forecasting literature	Feature-conditioned combination learns weights over a heterogeneous pool	Its features are properties of the series, computed once from history	A generalisation along one identified axis, stated as such

Fifty years of evidence says equal weighting is hard to beat, which is why it is the pre-registered primary benchmark rather than a convenience baseline.

Convergence

Literature, systems, CSE data, and forecasting methods point to the same missing mechanism.

The benchmark

Equal weighting is the formal pre-registered primary benchmark.

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Three themes, and what each one forces on the design

THEME 1 · THE MARKET

Thin trading is the condition here, not an edge case

Volatility clustering in three CSE studies, long memory across all three regimes, and illiquidity shocks depressing returns.

The conditioning variables are locally evidenced, not borrowed

THEME 2 · COMBINATION

Every necessary clause exists somewhere, in a different system

Analyst pipelines, feature-conditioned combination, continuous state conditioning, and explicit weight vectors over assets.

The gap is the conjunction, so the contribution is a generalisation

THEME 3 · EVALUATION

A strong minority evaluates well, and almost none can be rerun

Walk-forward, regime decomposition and cost sweeps appear. Of my 59 papers, 4 are reproducible from the paper alone.

Cached outputs across arms, a pre-registered margin, a released pipeline

Counts are of my own 59-paper corpus, not of the field. The classification rule that produced them is in the backup.

Design impact

Each thematic critique establishes an explicit architectural constraint.

Reproducibility

Only 4 of 59 literature papers offered standalone replication.

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Every alternative also produces weights. The question is what the weights may depend on

WHAT EXISTS

Multi-agent systems that combine analysts, mostly through a prompt. Influence fixed at design time.

WHAT IS MISSING

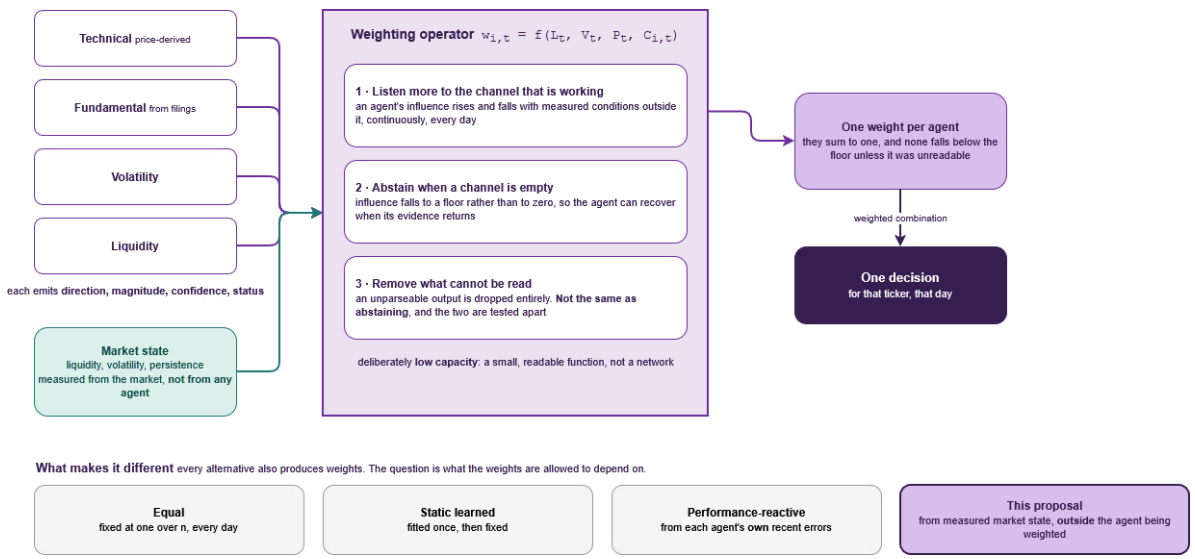
A weight that depends on conditions measured outside the agent being weighted, recomputed every step.

THE CONTRIBUTION

Attributable evidence on whether state-aware orchestration earns its place, positioned as a generalisation of feature-conditioned combination along one named axis.

Not a duplicated implementation. Three claims were withdrawn once I found papers that already did them, and the withdrawals are on the record.

Fig. 2.1 · Decision Flow Architecture



The weighting operator $w_{i,t} = f(L_t, V_t, P_t, C_{i,t})$ and decision flow across four disparate agents.

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No question is answerable yes or no. Each asks how

MAIN RESEARCH QUESTION

How should the influence of heterogeneous agents be adapted to measured market state, and under what conditions does adaptation earn its place?

Answered by five arms over byte-identical cached outputs, with one ablation per state input. Success: the median per-fold gain exceeds the interquartile range of the equal-weight arm across 16 folds, and clears HAC and bootstrap tests at 0.05.

SUPPORTING QUESTION 1

How should the system alter execution and the influence of price-derived agents when liquidity deteriorates?

Sensitivity sweep of weights against liquidity, with two ablations that separate trust from execution.

SUPPORTING QUESTION 2

How can confidence from unlike architectures be placed on a common scale?

Four structural mechanisms, each mapped through its own accuracy-at-threshold curve. If they do not land in one unit, that is the finding.

OBJECTIVES

Build the artefact, then evaluate it honestly.

Data layer, four agents behind one contract, state vector, operator, five-arm harness. Acceptance criteria A1 to A6. A null at the pre-registered margin is acceptable, decided in advance.

Evaluation rigour
Every research question ties directly to a falsifiable experiment with a pre-registered decision threshold.

Pre-registration
A null result is pre-registered as fully acceptable.

Each question has a pre-specified response if the answer is unfavourable, written before any run.

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The nearest system in each direction gets one clause right

Table 2.2 · Clause-by-Clause Baseline Comparison

System or baseline	What it gets right	Where it stops	What that forces on my design
Prompt-combined agent systems	Heterogeneous roles feeding one decision	They combine through free text, so no weight object exists	A common structured output contract has to come first. FR2
HedgeAgents	Optimisation over an explicit weight vector	The weights are over assets, so which method to trust cannot arise	The single-shared-decision clause, and calibration before any optimiser
MacroHFT	Continuous conditioning on market state, weight trace published	Six identical networks, so conditioning without heterogeneity	The heterogeneity clause, with entropy collapse as an observed failure mode
Feature-conditioned combination	Learned weights over a heterogeneous pool, closest in form	Features are properties of the series, computed once from history	Named as the lineage of this work rather than a rival
Pavithya et al. (2021), on the CSE	Two unlike models, one decision, on this market	A fixed 3:1 split taken from a score on an easier task	The motivating failure: a weight cannot come from an out-of-domain score

The proposal is compared against four alternatives over identical cached outputs, plus an oracle fitted on the test window held as a hindsight ceiling, not an achievable arm.

Lineage
Rather than claiming an ungrounded break, the project isolates where current state-of-the-art architectures stop.

Pavithya et al.
Direct CSE precedent: motivating the rejection of static cross-task weighting splits.

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The exposure is not confidentiality. It is false confidence

PRIMARY OPERATOR · THE RESEARCHER

Needs reproducible runs and comparisons that are actually fair.

Evidence: a published comparison gave its baseline less scaffolding than the proposed system. **Design:** byte-identical cached outputs across all five arms, enforced architecturally.

MOTIVATING BENEFICIARY · THE INVESTMENT DECISION-MAKER

Needs a decision that does not carry confident influence from an empty channel.

Evidence: five documented cases of reliability varying unnoticed, including a 10.36-point loss. **Design:** the abstention channel and the confidence gate.

THE EXCHANGE AND ITS DATA REGIME

Licensed data must be used within its terms.

Evidence: the purchased set carries 46 securities absent from the API, a 12.8 per cent gap. **Design:** official licensed extraction as the base panel, licence recorded per row.

WIDER EXPOSURE · RETAIL PARTICIPANTS

Should not be exposed to a system that assumes away the cost they would pay.

Evidence: institutional ownership at 66.1 per cent against 5.7 per cent insider. **Design:** cost inside the allocation objective, and the artefact framed as a prototype rather than a recommender.

A well-formed directional signal produced from a channel that is empty today is the failure this system is designed to make visible.

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REQUIREMENTS ENGINEERING · CRITERION 3, 20 PER CENT

Every requirement carries the evidence that forced it

Table 3.1 · Traceability Sample

ID · Requirement · Type · Source or rationale · Acceptance evidence

FR1 · The data layer shall refuse any row past the simulation clock, raising rather than warning · Functional
Two documented point-in-time failures on this market
Cursor-violation test and an exception log

FR2 · Every agent shall emit the common output contract, all six fields · Functional
Prompt-combined systems in which no weight object exists
Contract conformance across all four agents

FR10a · The system shall emit the weight entropy series and the liquidity sensitivity · Functional
Acceptance criteria A2 and A3 are defined on these two quantities
Entropy series and sensitivity sweep plot

NFR7 · Outputs, state and weights shall be logged one row per ticker-date · Quality
An explanation that was not recorded is not an explanation
A sampled decision rebuilt end to end from the log

DR10 · Ex-date reliability shall be carried as a field, with inferred dates flagged · Data
No documented date in the registers is an ex-date; 259 are approximate
ex_date_basis populated; affected days flagged

B2 · No fifth agent, including a macroeconomic agent · Boundary
Rejected on evidence: adjusted R-squared 0.189 across five macro variables
Scope statement carrying the measurement

43 requirements: 11 functional, 7 non-functional, 10 data, 7 model and AI, 8 boundary. Exactly three are Should. Six shown here; the full traceability matrix is backup slide B3.

Requirements tally — 43 total requirements across 5 categories, with full source-to-test traceability.

Boundary rigor — B2 excludes macro signals based on empirical evidence ($R^2 = 0.189$). Slide 10

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